

SERENITY

Smart Emotion Recognition & Neural Intervention Technology

Group Members:

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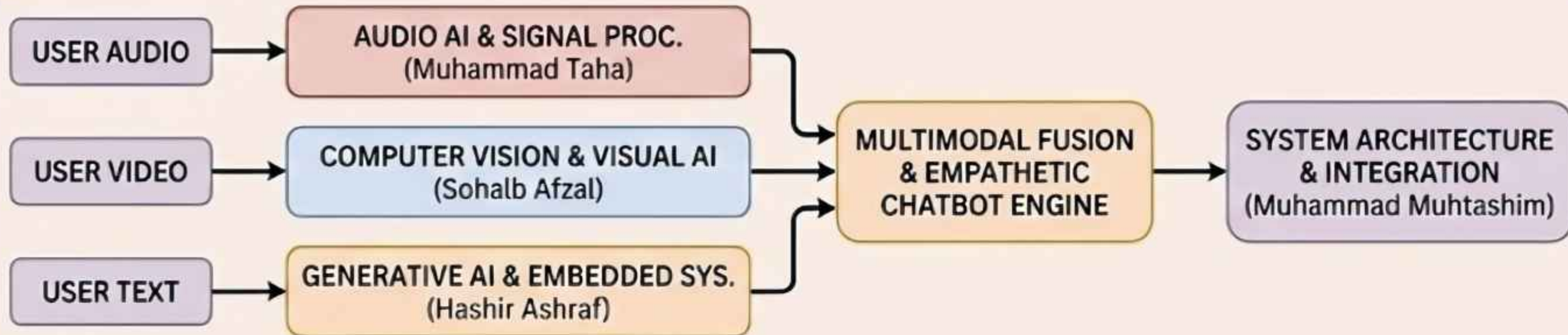
Dr Farhan Hussain

Industrial Partner:

Chosun University South Korea

PROPOSED SOLUTION WITH TIMELINE

AI-BASED MENTAL CARE SYSTEM OVERVIEW & WORK DIVISION



PROJECT TIMELINE (PHASES 1-5)



LITERATURE REVIEW AND PROPOSED SOLUTION

Problem Context & Emotion Recognition

- Mental health monitoring needs objective & real-time emotion analysis.
- Single-modality systems are limited & delayed.
- Literature supports continuous emotion-aware systems.

Mini Table (Simplified):

Modality	What Literature Shows
Face	CNNs (ResNet) → strong spatial features
Speech	CNN + Bi-LSTM → spectral + temporal cues
Text	Transformers → contextual emotion

📖 Ekman, Trigeorgis et al., Vaswani et al.

Model Engineering (Literature-Guided)

1 Facial Emotion Recognition

- 🧠 **ResNet (CNN)**
 - Residual connections → avoids vanishing gradients
 - Proven stability in FER literature

2 Speech Emotion Recognition

- 🎤 **CNN + Bi-LSTM**
 - CNN → local spectral patterns
 - Bi-LSTM → temporal dependencies

3 Attention Mechanism

- 🔍 **Multi-Head Attention**
 - Emphasizes MFCC, ZCR, RMS
 - Improves accuracy & F1-score

Multimodal Fusion

- 🔗 **Late Fusion > Early Fusion**
 - Decision-level robustness
 - Literature-supported

📖 Poria et al., Multimodal Sentiment Analysis.

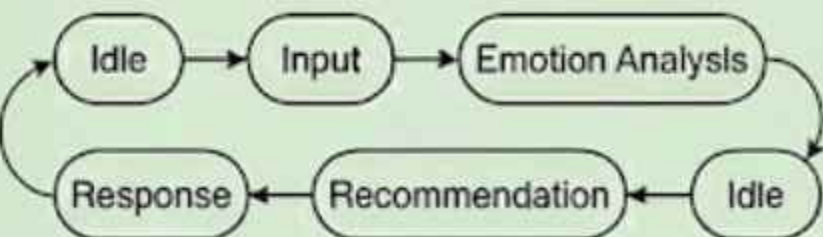
Research Gaps & Proposed Solution

Gaps in Literature

- Single-modality focus
- Cloud-dependent systems
- No real-time deployment
- Limited practical mental-care output

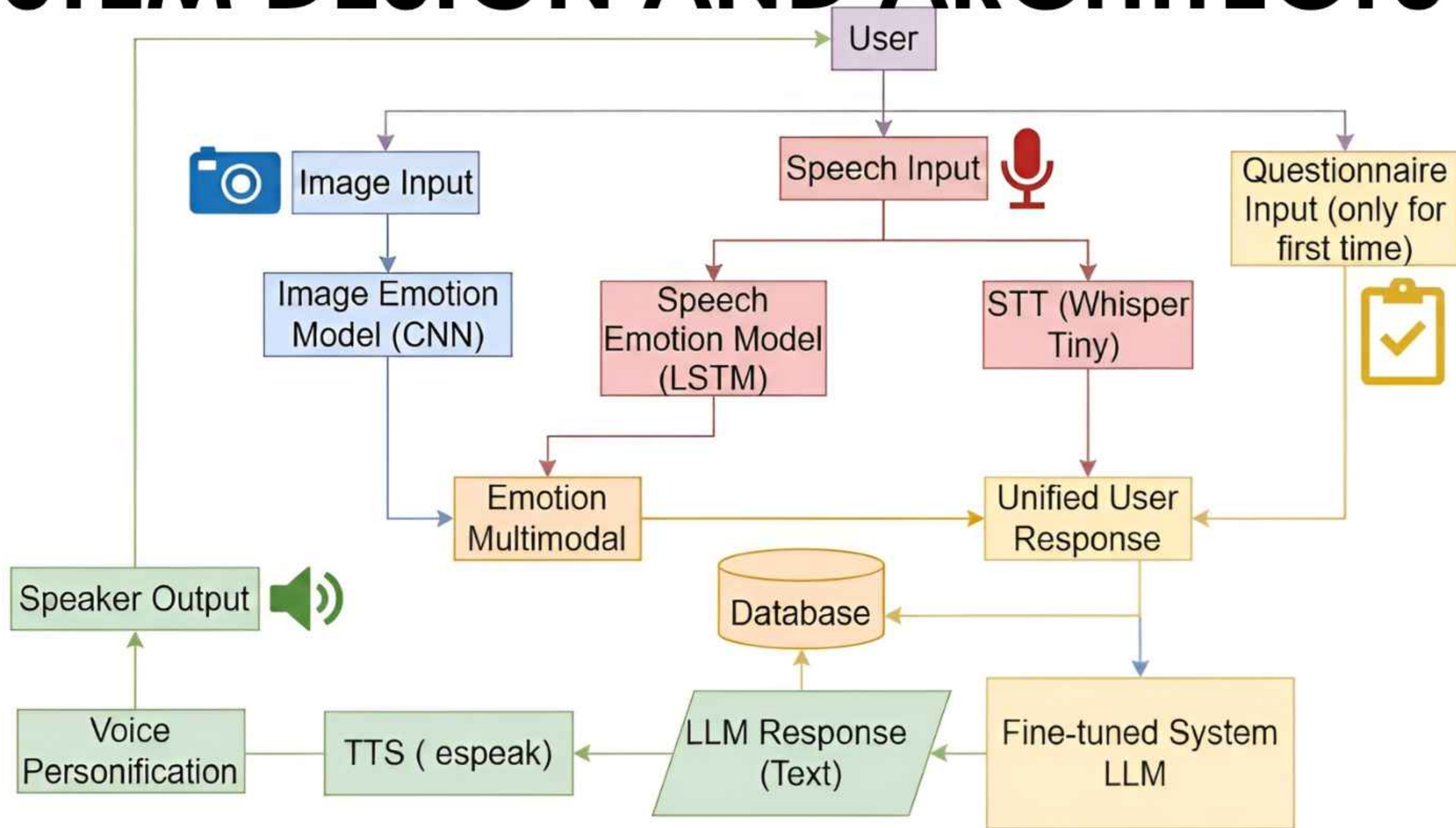
Our Contribution

- Multimodal Emotion Recognition (Face + Speech + Text)
- Hybrid Deep Models (CNN + Bi-LSTM + Attention)
- Real-Time Edge Deployment (Raspberry Pi 5)
- Emotion-Aware State Machine



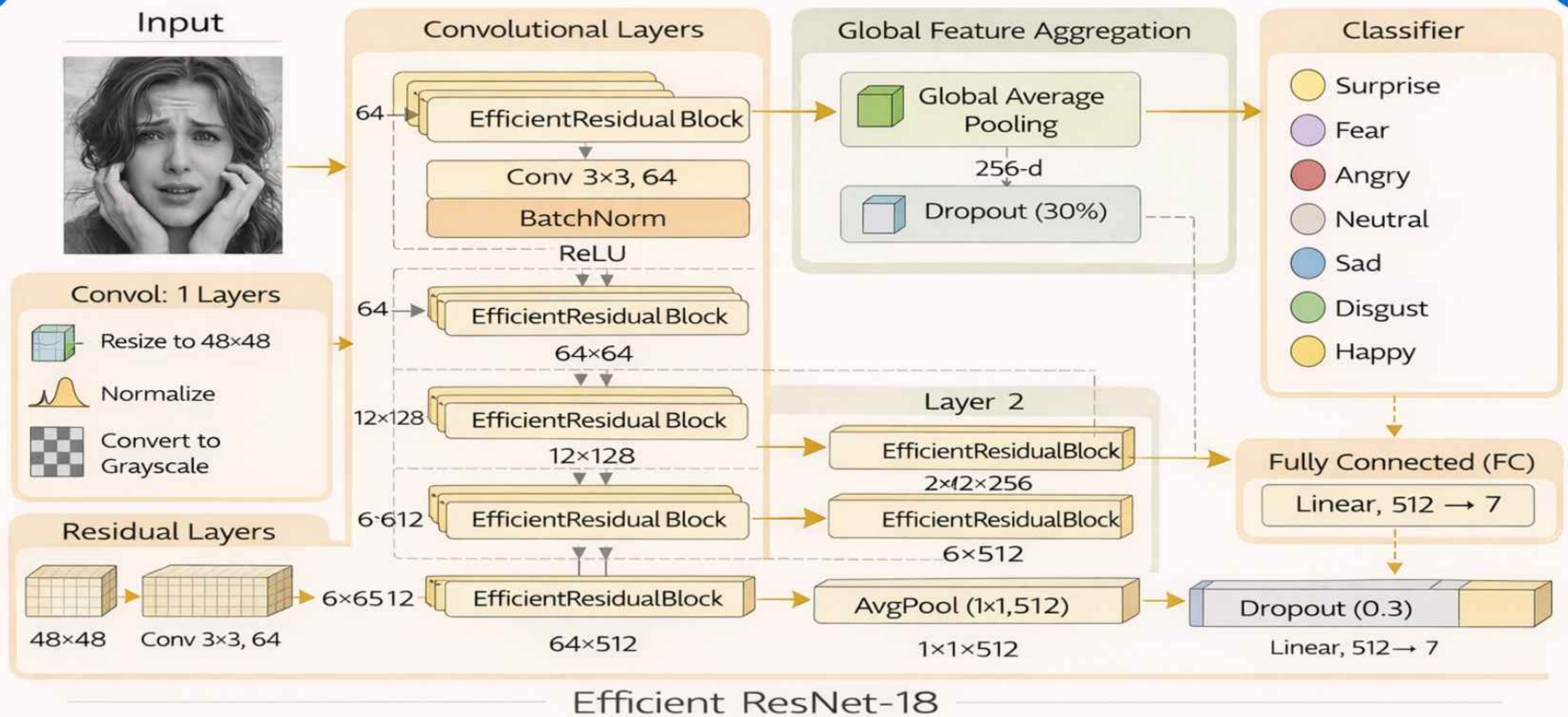
"Literature-consistent multimodal emotion recognition with real-time edge deployment."

SYSTEM DESIGN AND ARCHITECTURE



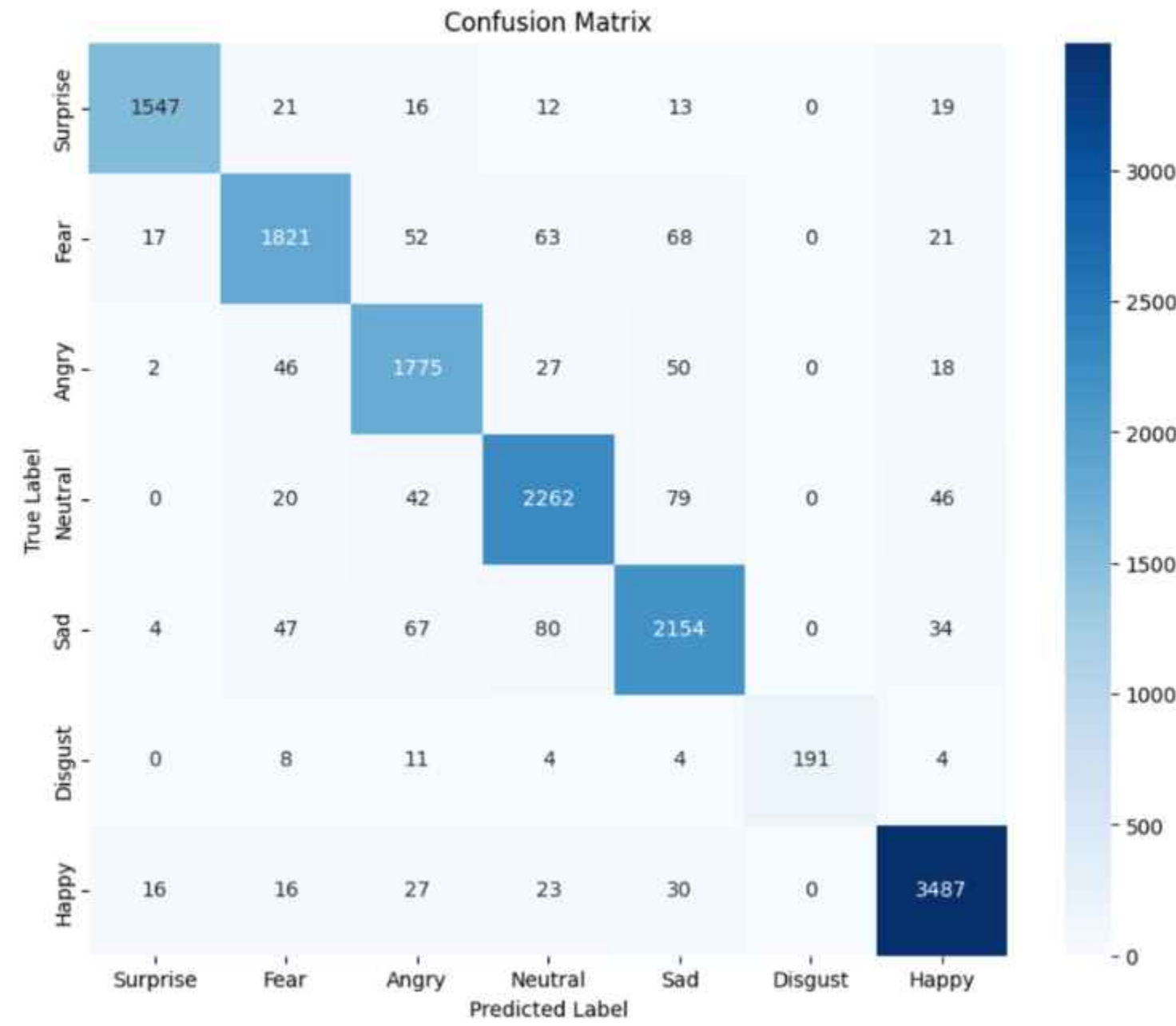
Overall Workflow: Multimodal Sentimental Analysis and LLM-Driven Mental Care

MACRO FACIAL EXPRESSION



MACRO FACIAL EXPRESSION

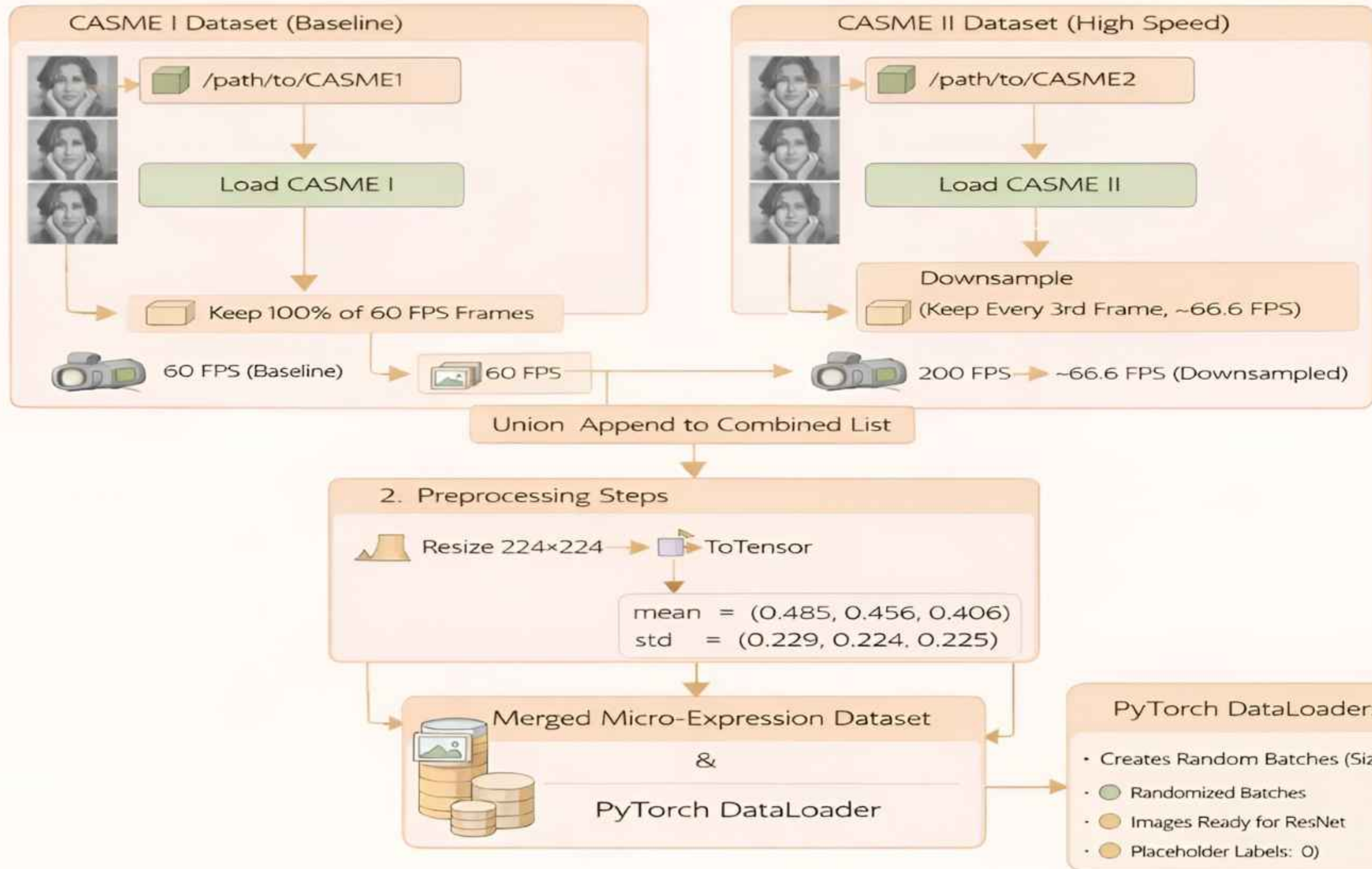
RESULTS



	precision	recall	f1-score	support
Surprise	0.98	0.95	0.96	1628
Fear	0.92	0.89	0.91	2042
Angry	0.89	0.93	0.91	1918
Neutral	0.92	0.92	0.92	2449
Sad	0.90	0.90	0.90	2386
Disgust	1.00	0.86	0.92	222
Happy	0.96	0.97	0.96	3599
accuracy			0.93	14244
macro avg	0.94	0.92	0.93	14244
weighted avg	0.93	0.93	0.93	14244

MICRO FACIAL EXPRESSION

Merging & Preprocessing CASME I and CASME II

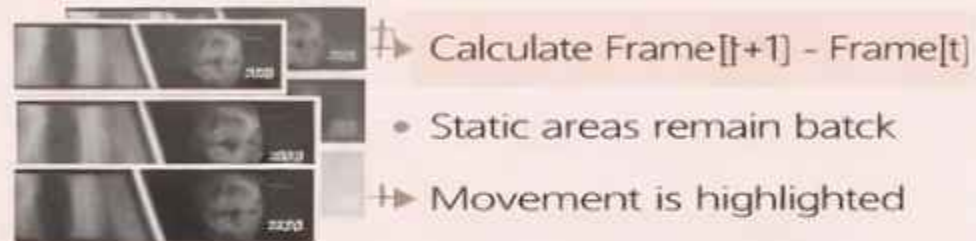


MICRO FACIAL EXPRESSION

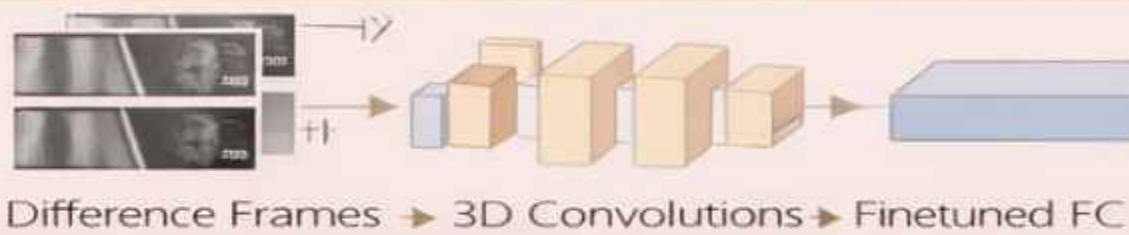
3D Convolutional Neural Network (3D-CNN) for Micro-Expression Recognition

Unlike 2D-CNNs, this approach analyzes video sequences to detect subtle facial muscle movements over time.

1. The Data Pipeline: "Difference" Frames



2. The Model Architecture: ResNet3D-18



3. Smart Training Strategy

➕ Subject-Independent Data Split

- Splits by Subject ID, not by file count.
- Ensures model is tested on unseen people.



▶ Handling Imbalance (Weighted Sampling)

- Weighted Sampler for Class Imbalance
- Ensures Rare Expressions Are Seen Frequently

4. The Execution Flow

📁 Label Matching (Excel)

- Scans Directories for Emotion Excel files

- Stack 16 video clips (112x112) into a tensor (Batch, Channels, T, H, W)

📁 Batch Creation

- Batch, Channels, T, H, W

📁 Batch Creation (16 Clips)

3D ResNet → CrossEntropy

(Batch, Channels, T, H, W)

🔄 Forward Pass

- 3D ResNet → CrossEntropy Loss

📄 Validation

Epoch End

Saving Best Model

MICRO FACIAL EXPRESSION

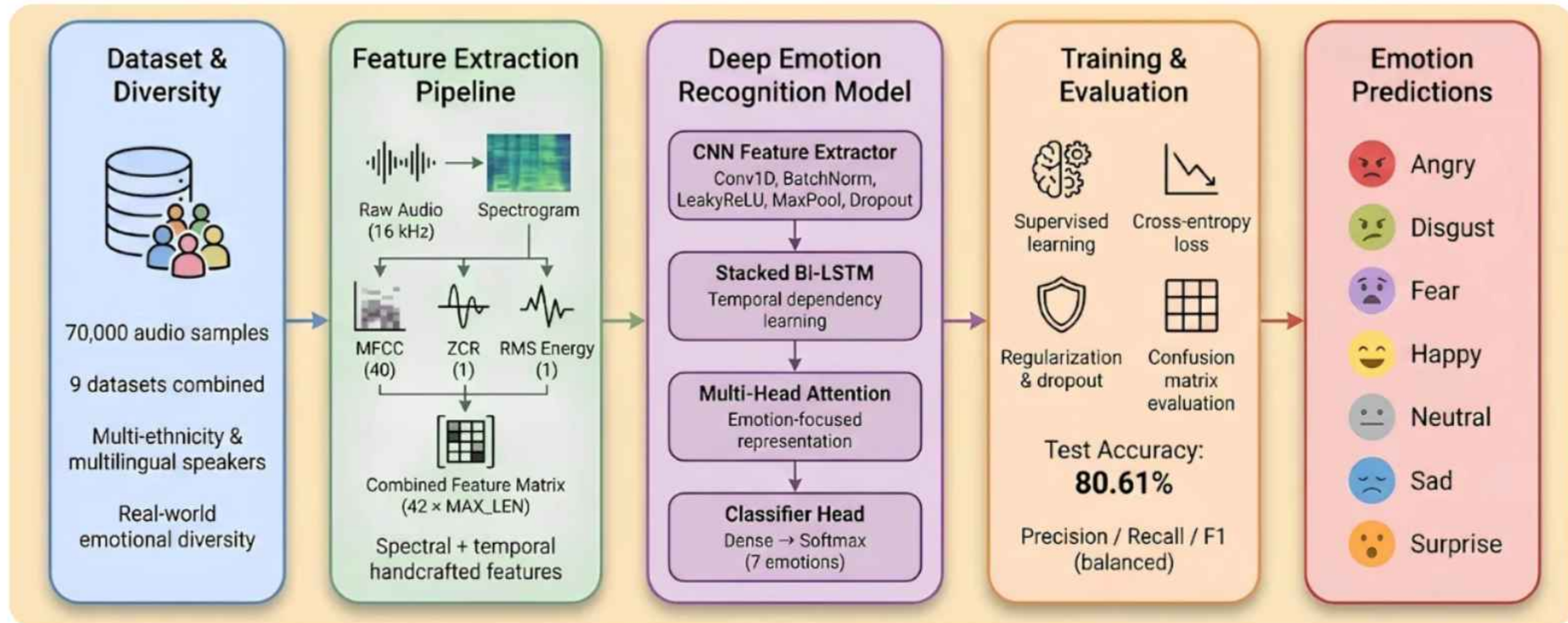
Best Validation Accuracy: 57.7%

	precision	recall	f1-score	support
happiness	0.00	0.00	0.00	6
happy	0.25	0.05	0.09	19
amusement	0.00	0.00	0.00	3
disgust	0.00	0.00	0.00	0
surprise	0.47	0.92	0.62	24
accuracy			0.44	52
macro avg	0.14	0.19	0.14	52
weighted avg	0.31	0.44	0.32	52

CHALLENGES AND SOLUTION IN MICRO EXPRESSION DATASET

- CASME I and II are very small, therefore we are using CAS(ME)3 dataset that overcomes all the issues that the previous datasets pose.
- But CAS(ME)3 has one problem, its 1.4 TB in size. So We are downloading this dataset in chunks each on individual laptops

SPEECH EMOTION RECOGNITION

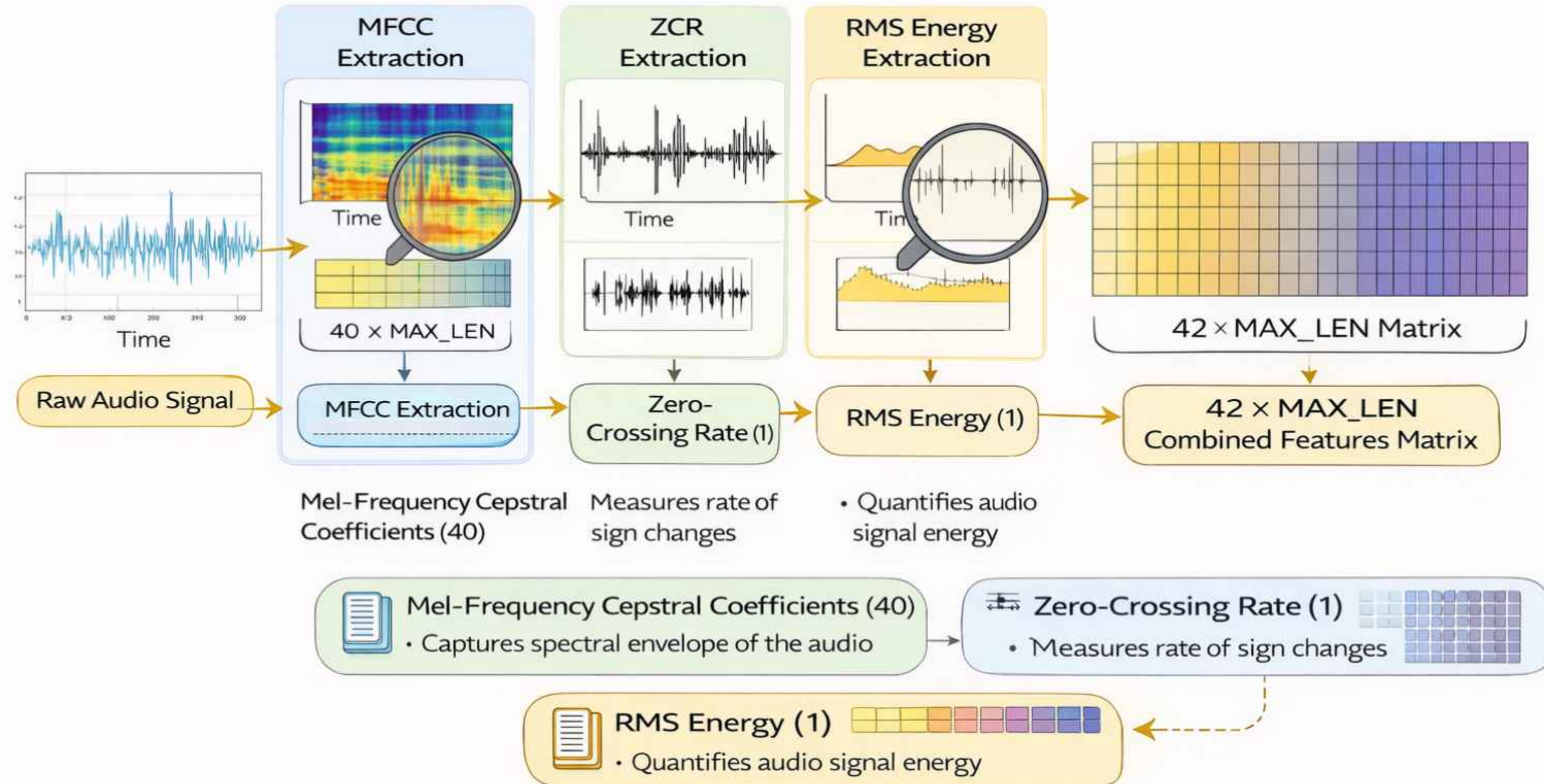


Training Datasets

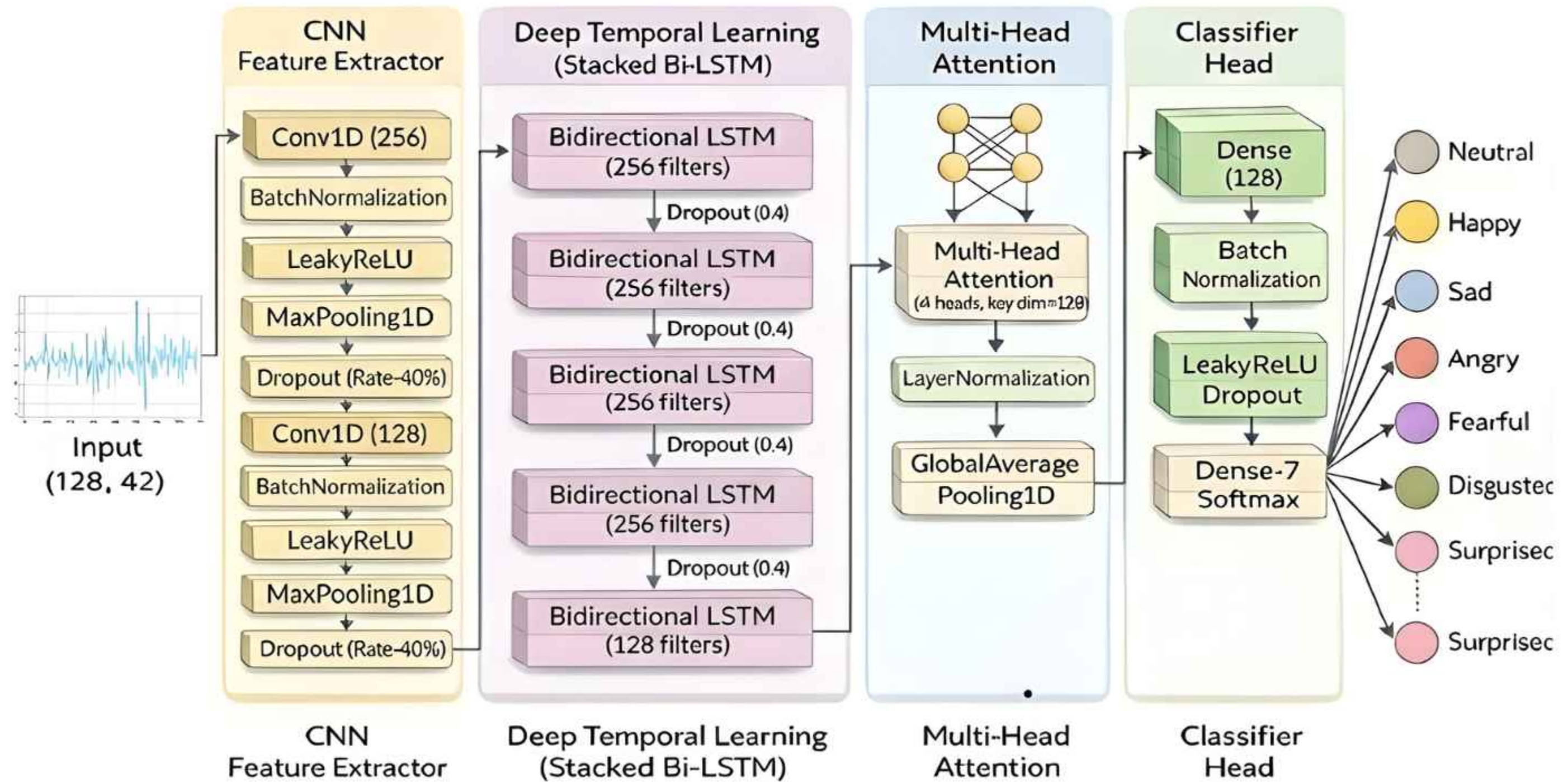
RAVDESS TESS CREMA-D SAVEE IEMOCAP JL Corpus
ASVP-ESD ESD EmoV-DB

SPEECH EMOTION RECOGNITION

FEATURES EXTRACTION



SPEECH EMOTION RECOGNITION MODEL ARCHITECTURE



SPEECH EMOTION RECOGNITION MODEL

RESULTS

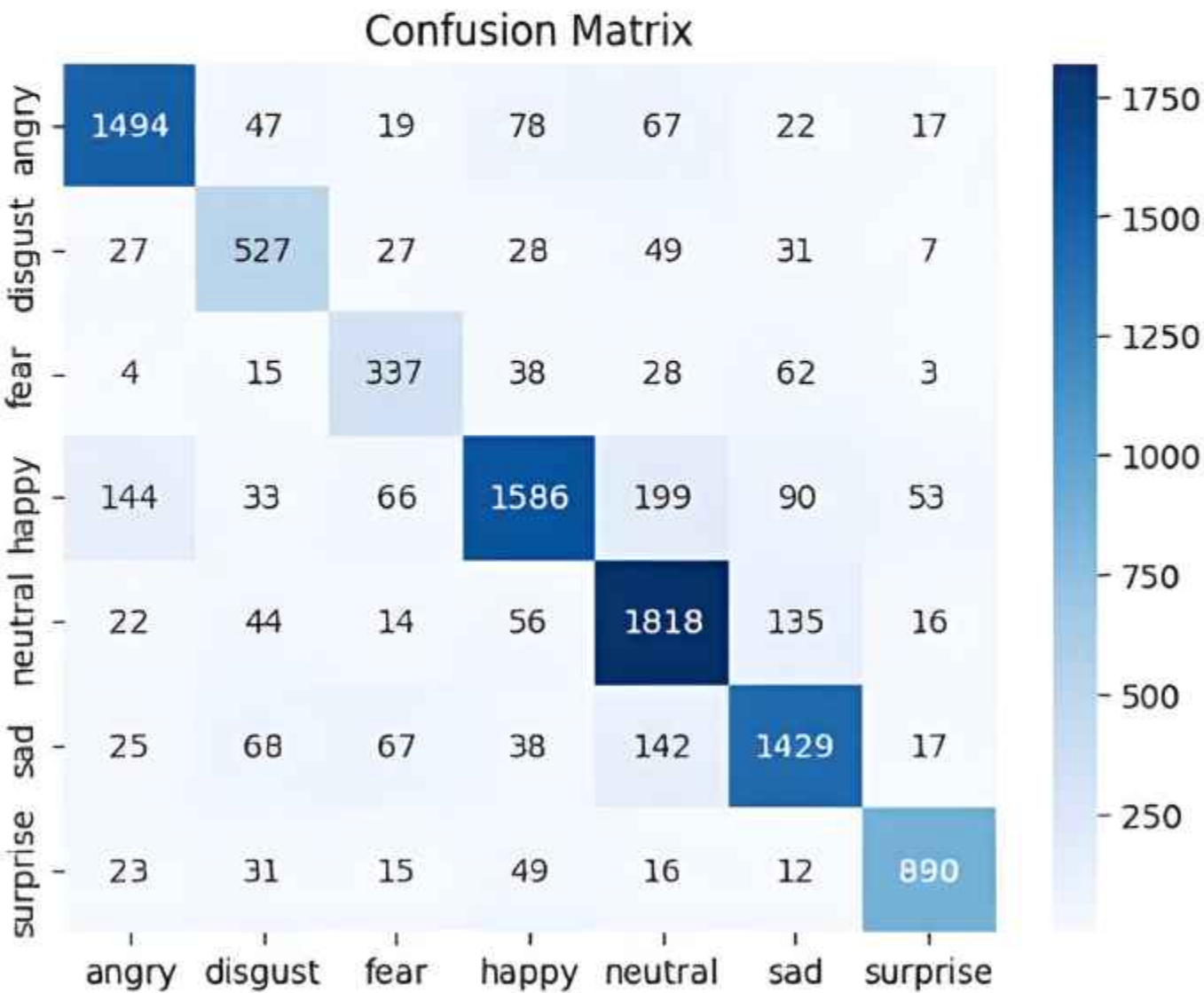
--- Final Test Evaluation ---

314/314 6s 10ms/step - accuracy: 0.8059 - loss: 0.7499

Test Accuracy: 80.61%

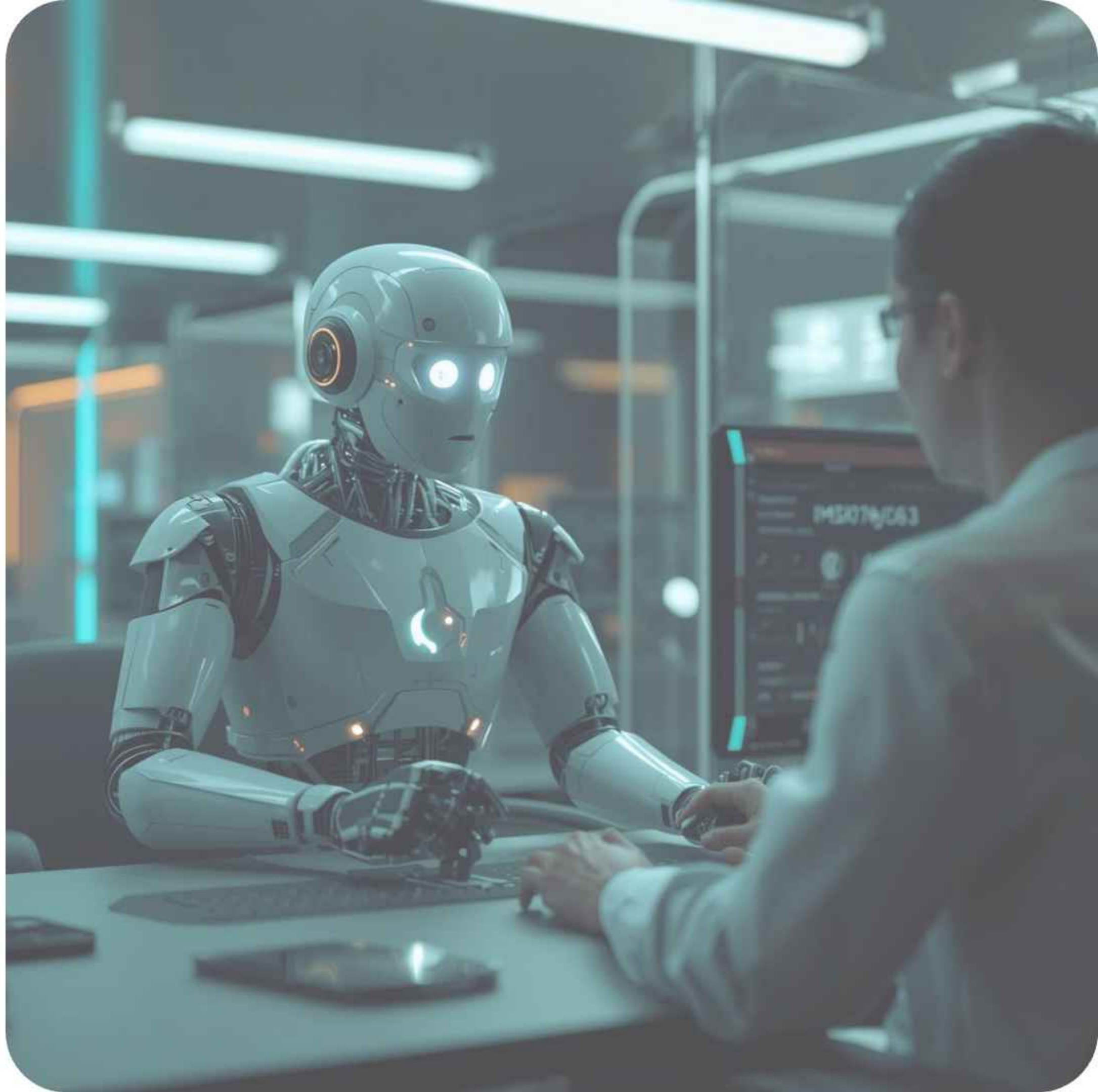
314/314 5s 11ms/step

--- Detailed Report ---

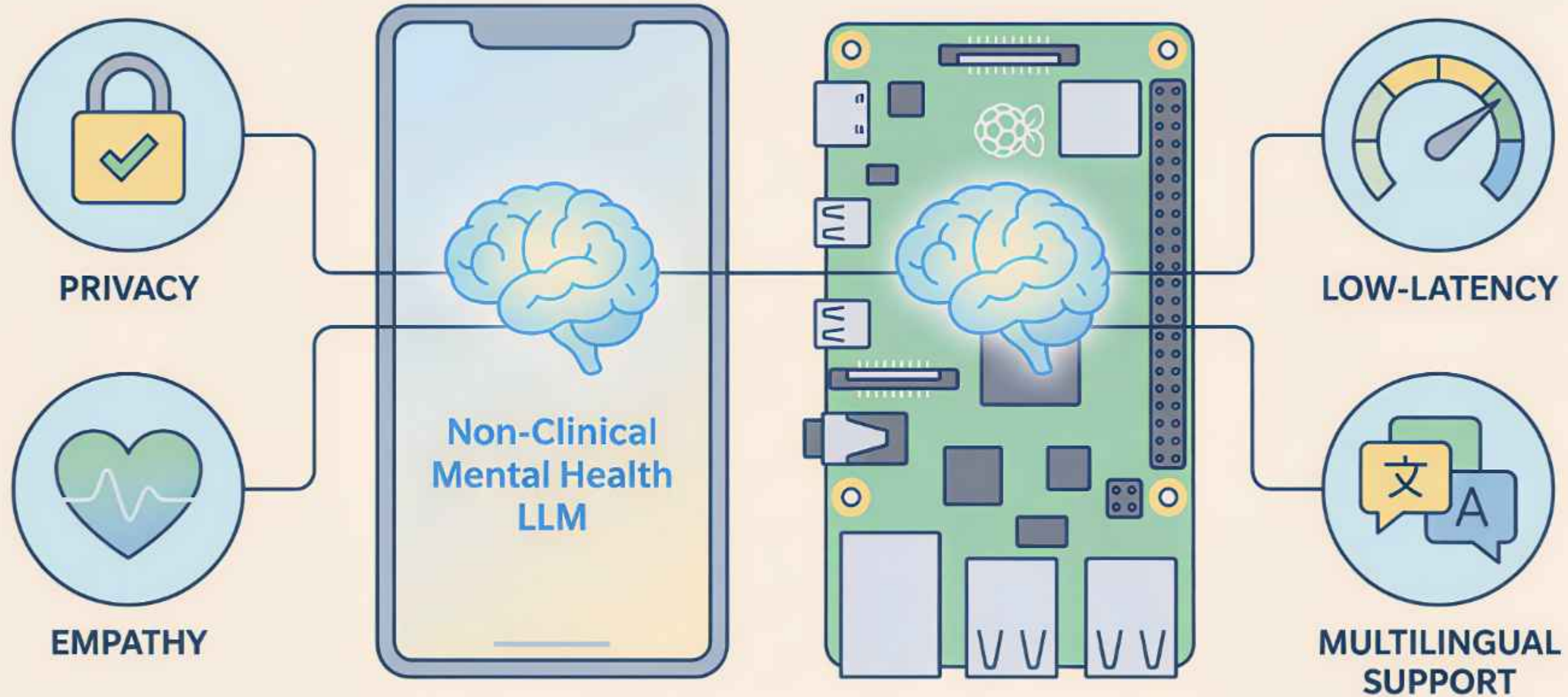


	precision	recall	f1-score	support
angry	0.86	0.86	0.86	1744
disgust	0.69	0.76	0.72	696
fear	0.62	0.69	0.65	487
happy	0.85	0.73	0.78	2171
neutral	0.78	0.86	0.82	2105
sad	0.80	0.80	0.80	1786
surprise	0.89	0.86	0.87	1036
accuracy			0.81	10025
macro avg	0.78	0.79	0.79	10025
weighted avg	0.81	0.81	0.81	10025

LLM FINETUNING

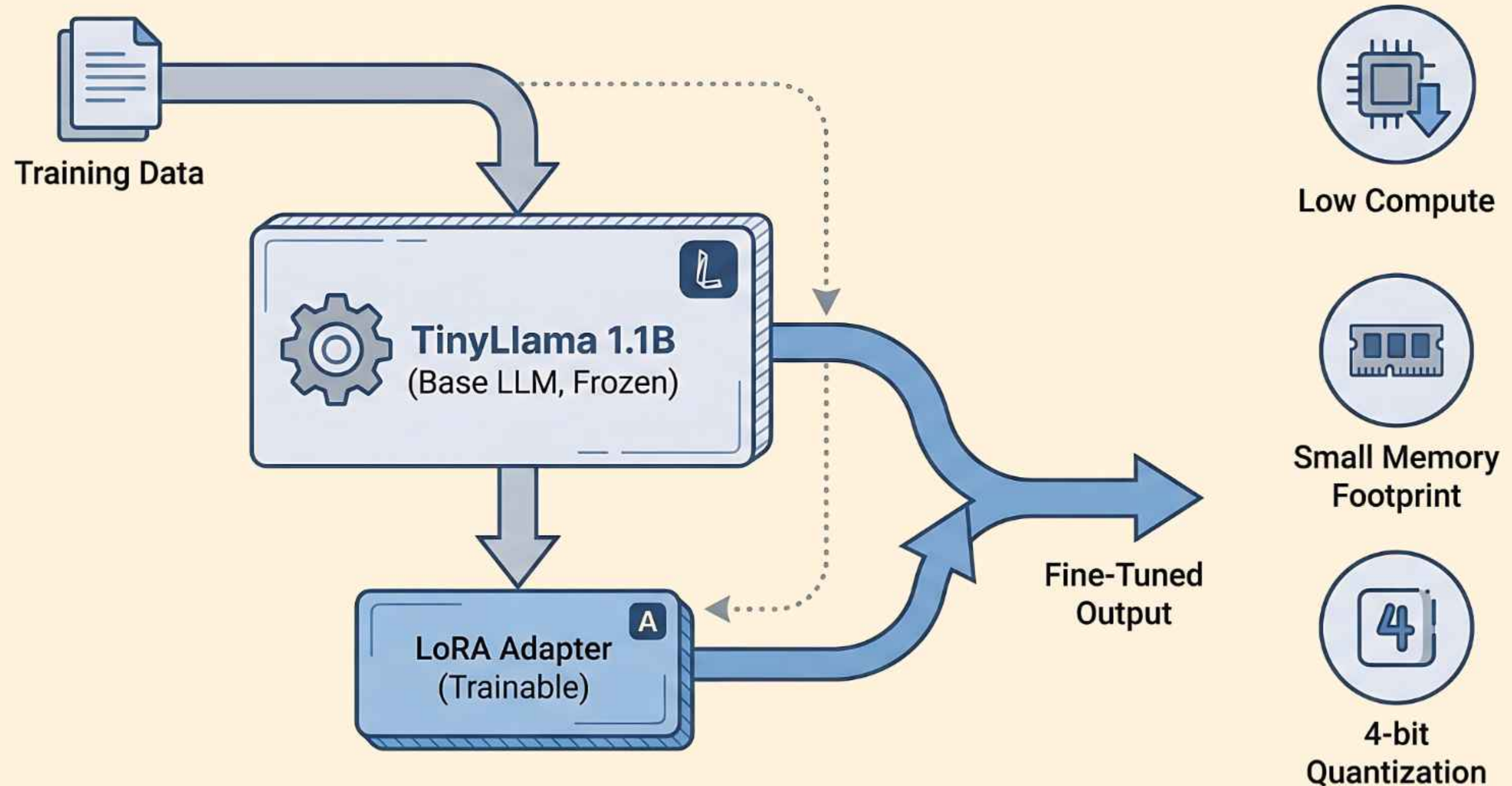


CONCEPT & OBJECTIVES

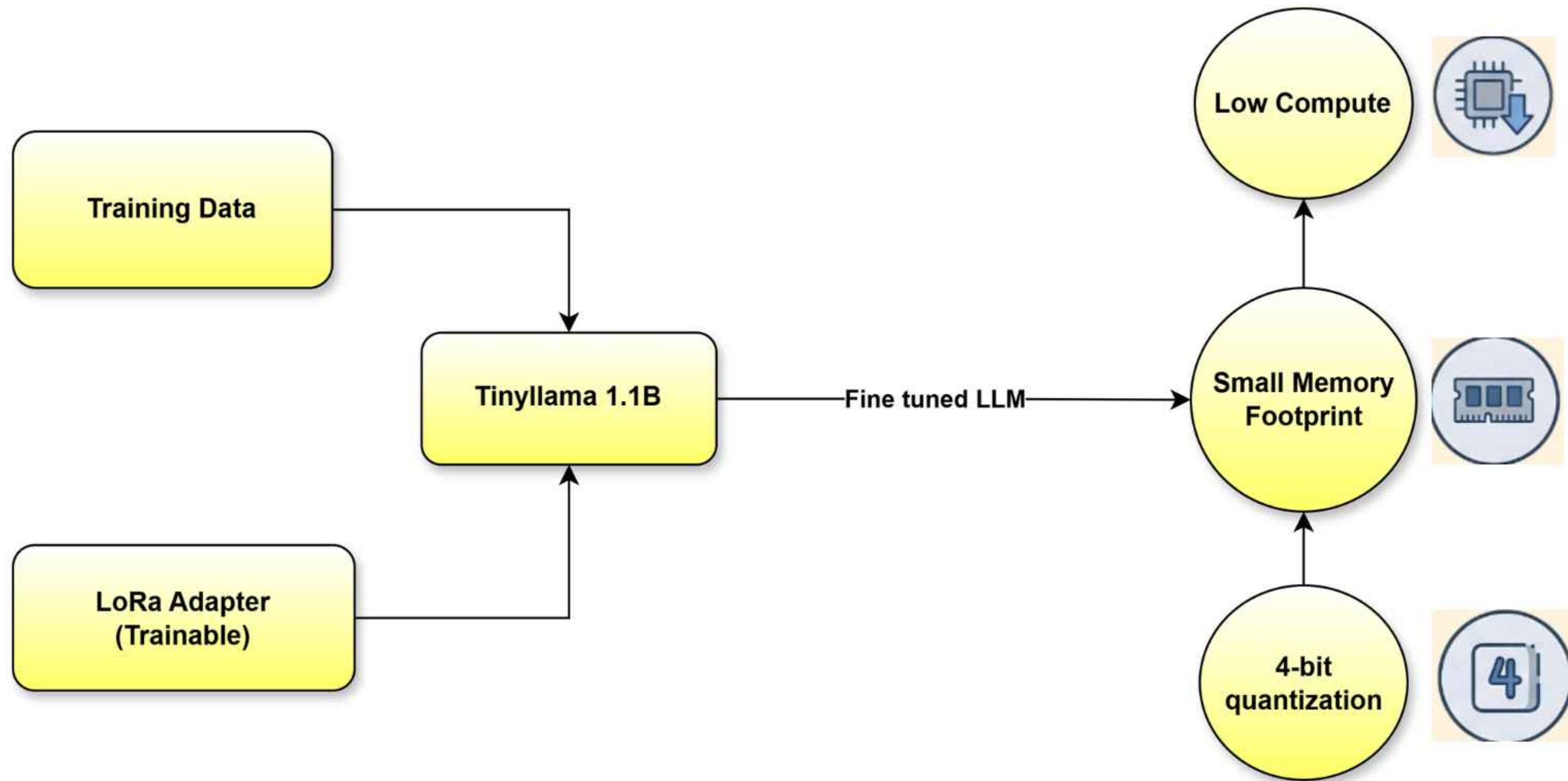


**On-Device Non-Clinical Mental Health
AI Assistant: A Privacy-First Approach**

MODEL ARCHITECTURE & TRAINING

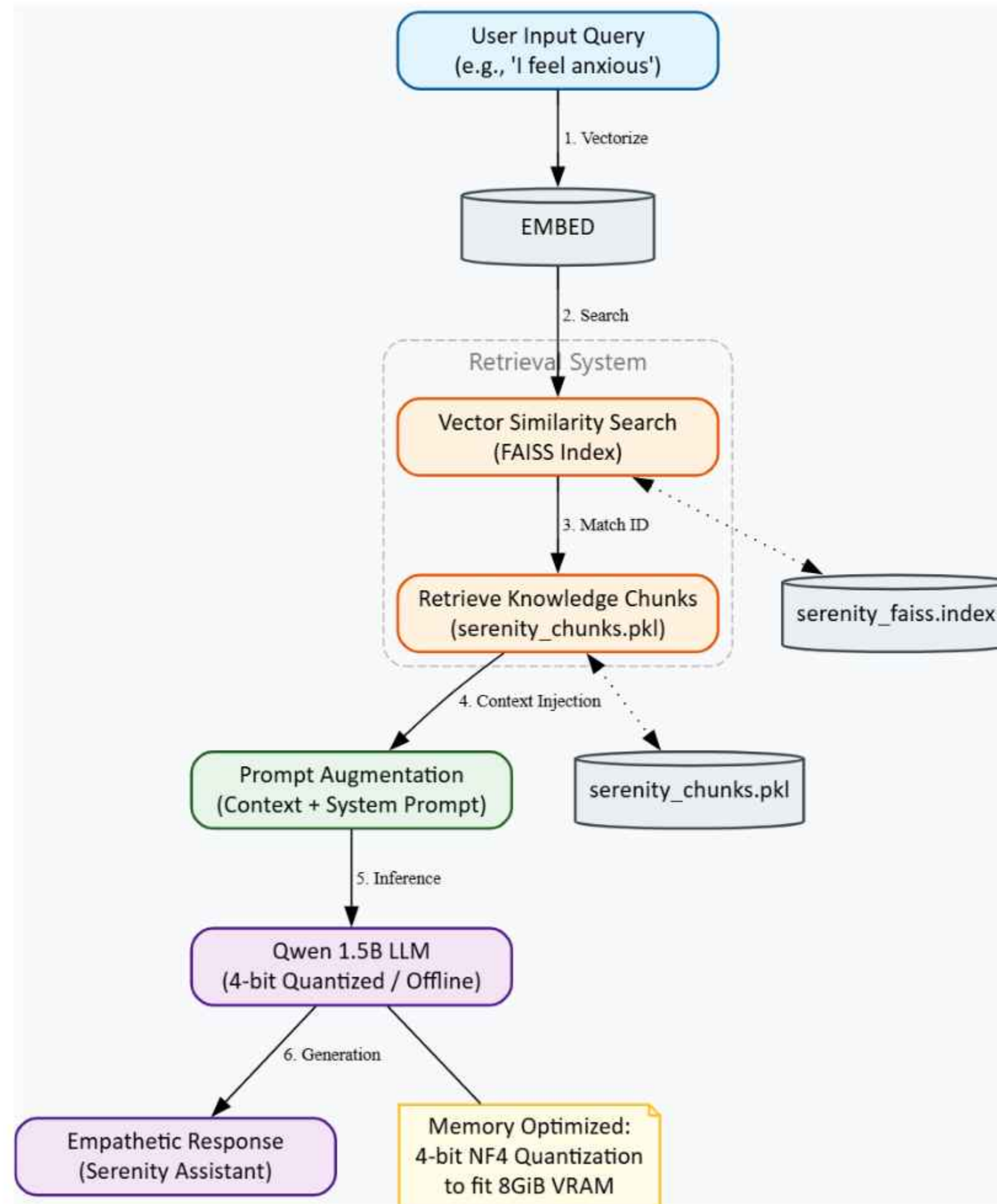


INITIAL APPROACH FINE-TUNING TINYLLAMA



Datasets used: Emphatic dialogues, Counsel Chat, DIAC WOZ

IMPROVED APPROACH



PREPROCESSING PIPELINE

DATA SOURCES



DAIC-WOZ
(Clinical Interviews)

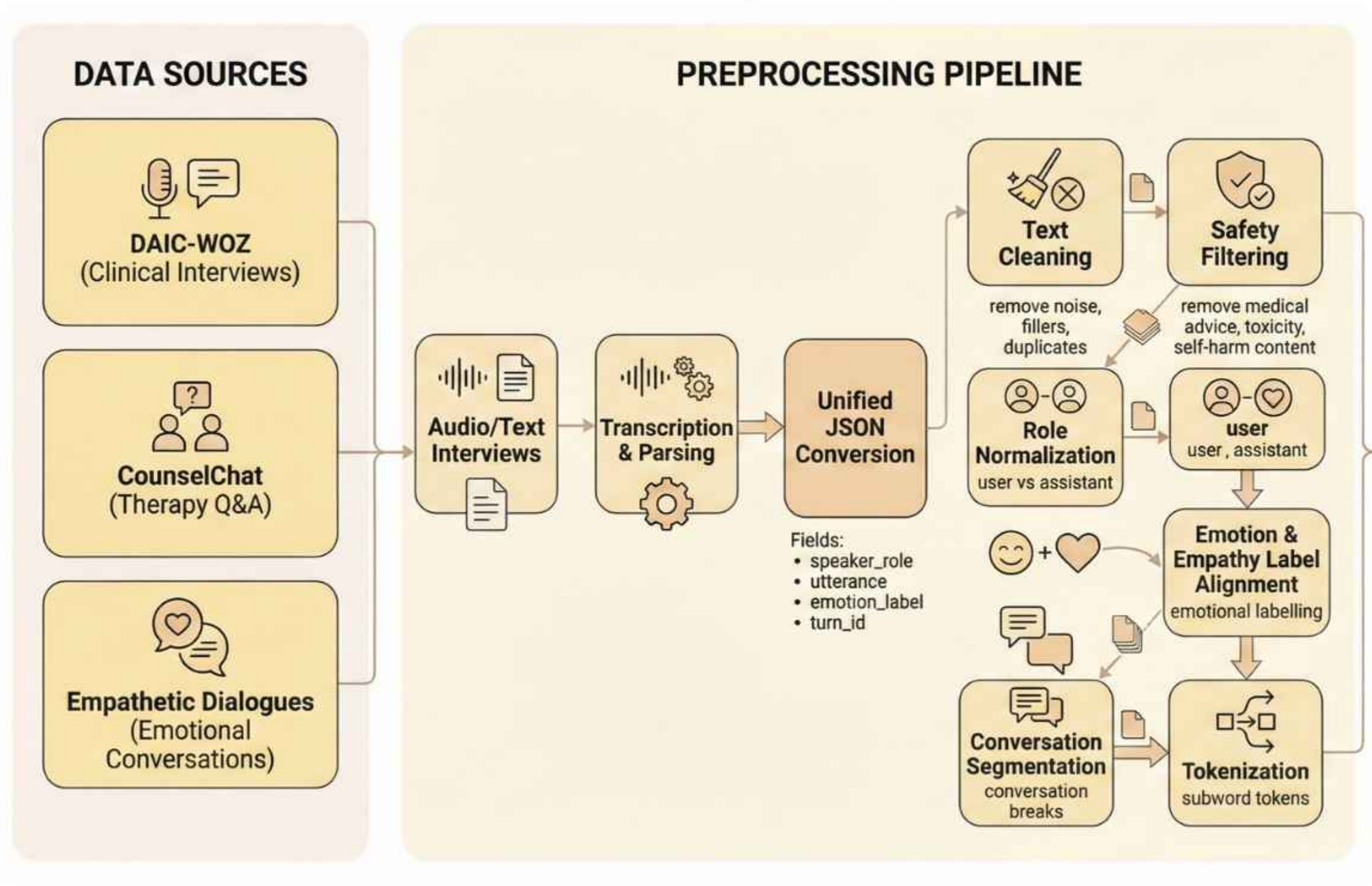


CounselChat
(Therapy Q&A)

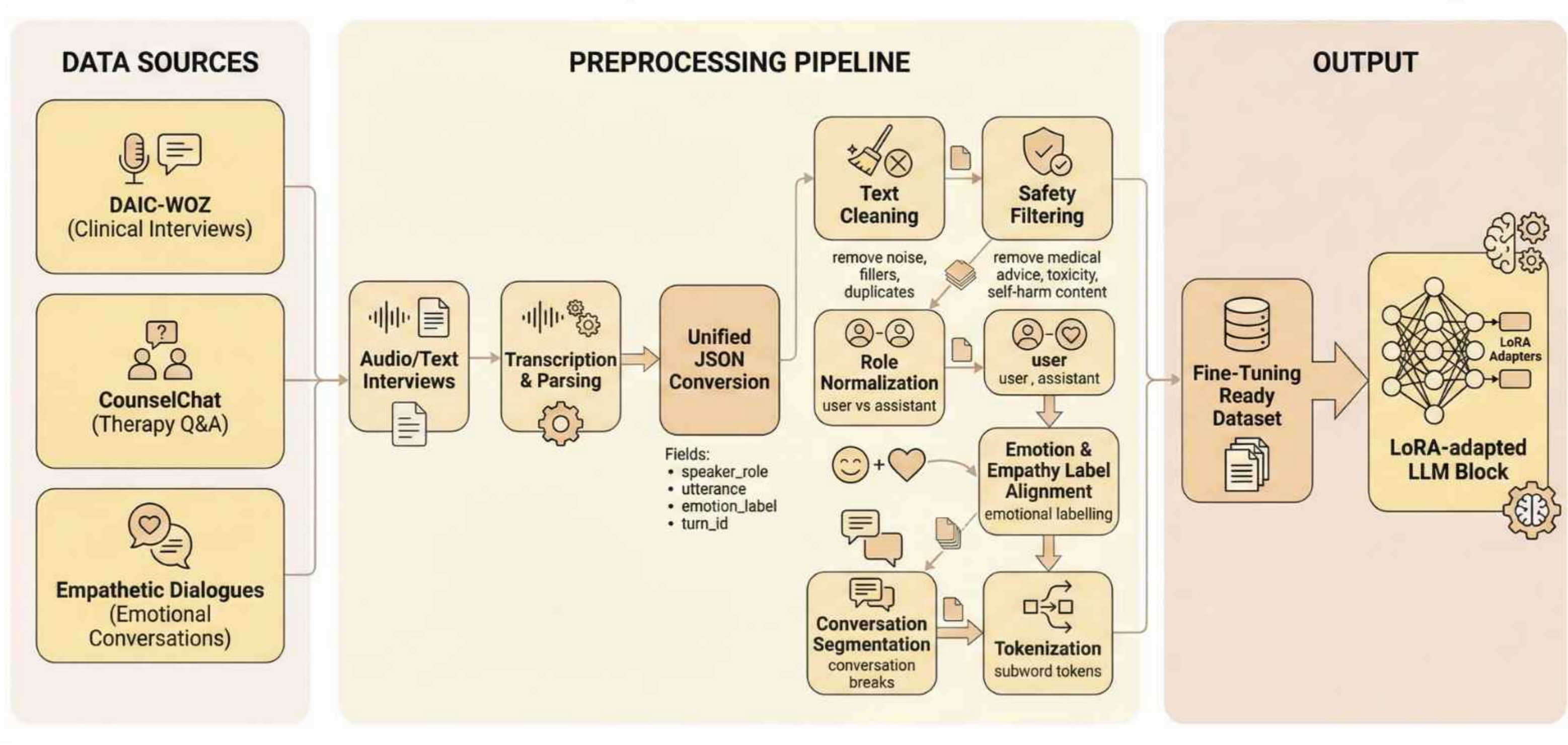


Empathetic Dialogues
(Emotional
Conversations)

PREPROCESSING PIPELINE



PREPROCESSING PIPELINE

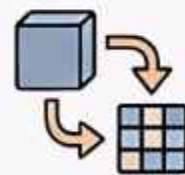


LLM FINETUNING PIPELINE

Step 1: Base Load

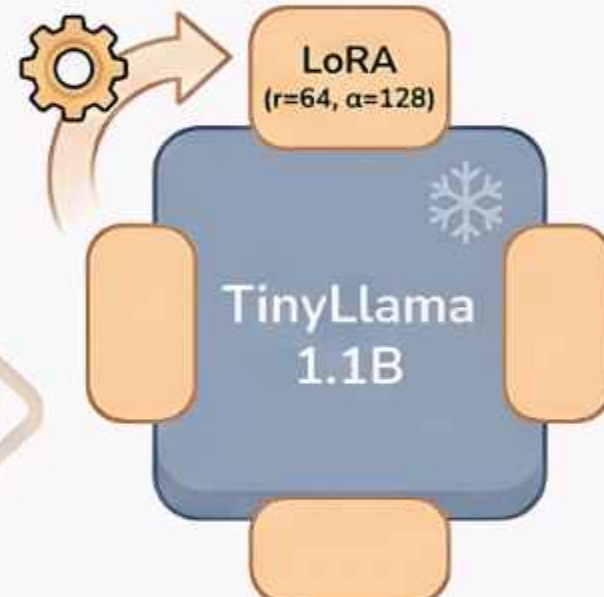


Loading
Weights



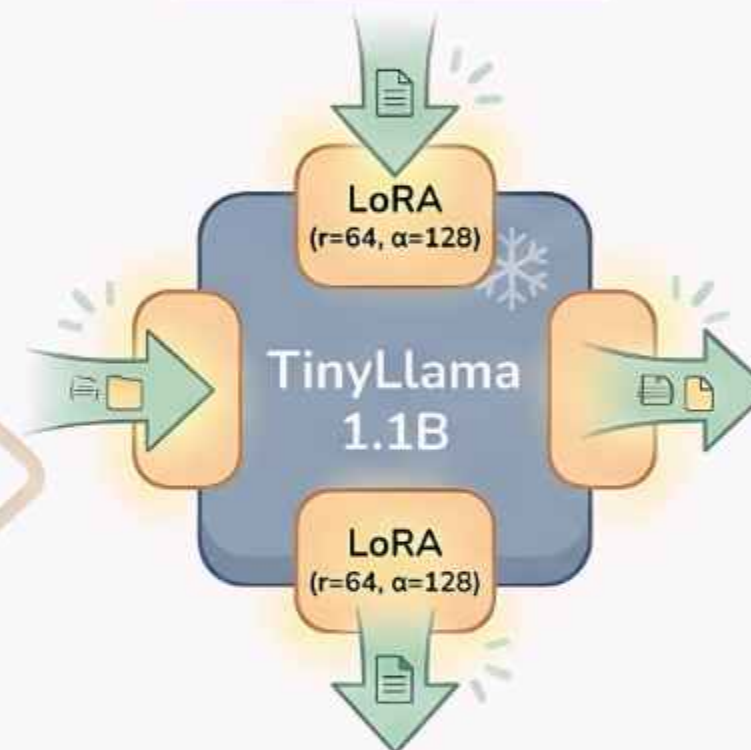
Quantization

Step 2: LoRA Setup

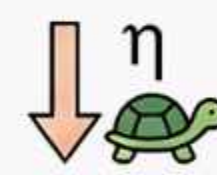


Parameter-Efficient
Adaptation

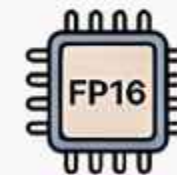
Step 3: Training



Supervised
Fine-Tuning

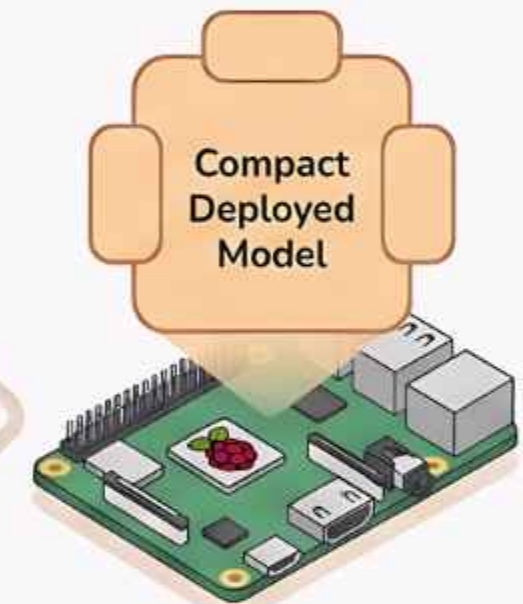


Low Learning
Rate



FP16
Precision

Step 4: Deploy

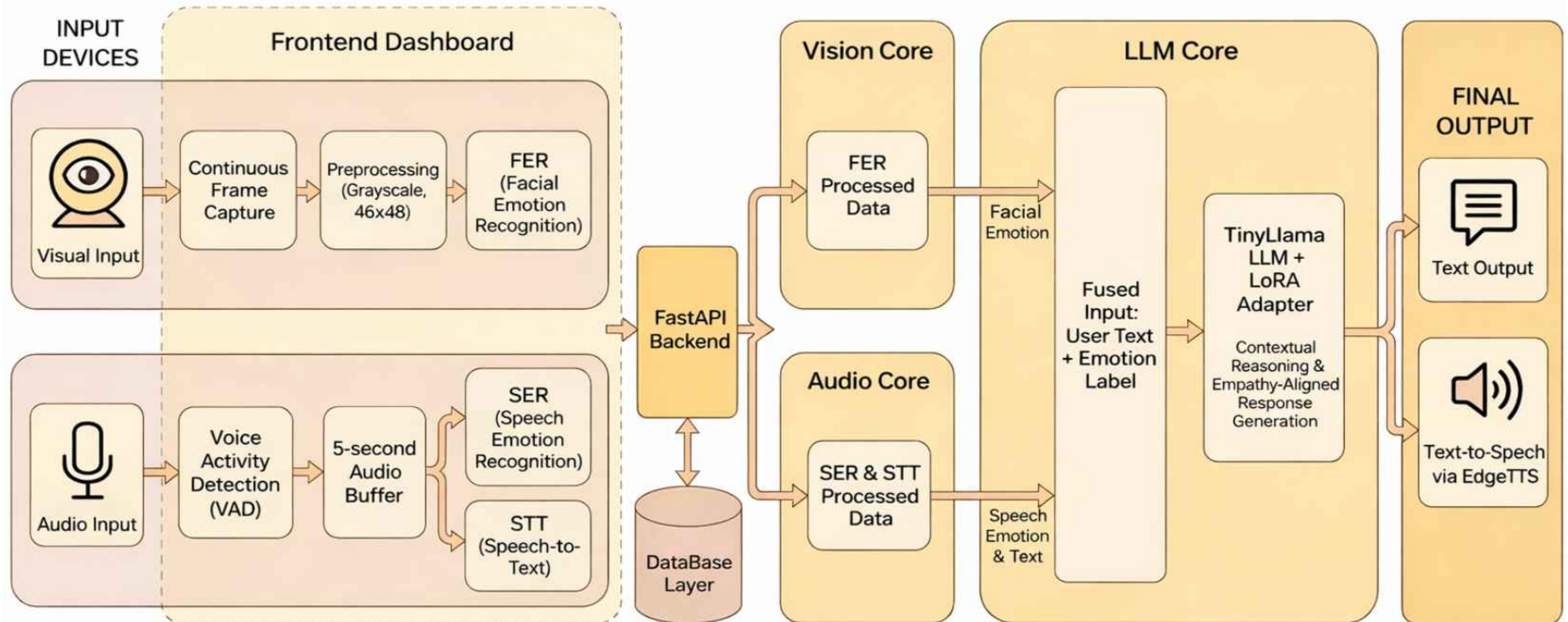


Safety
Filters



Ethical
Guards

DATABASE-LLM INTEGRATION



DATABASE: DESIGN & MANAGEMENT

Designed Central Database



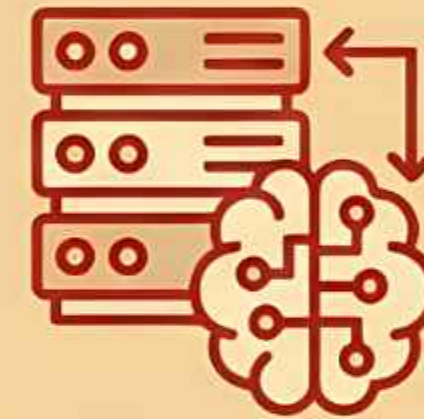
- Designed central emotion & user database.

Stored Content



- **Stored:**
- User profiles & preferences.
- Past emotional states (facial + speech).
- Conversation history.

Scalability & Importance



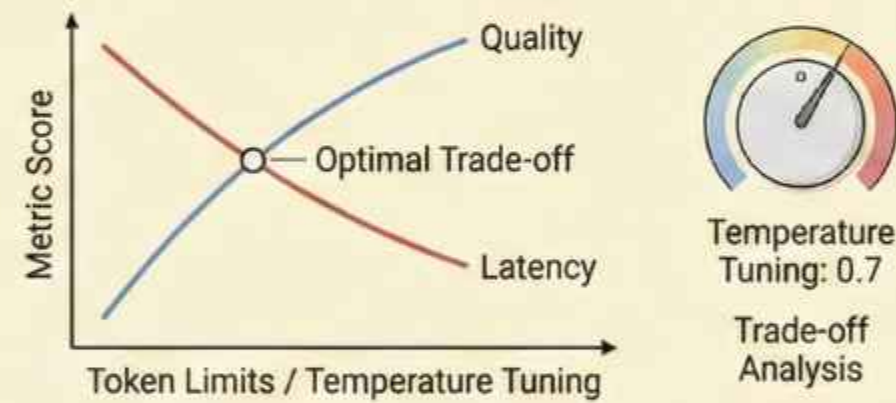
- Ensured scalable & organized data handling.
- **Importance:** Database acts as the “memory backbone,” allowing continuity in therapy-like sessions.

INVESTIGATION & EXPLORATION

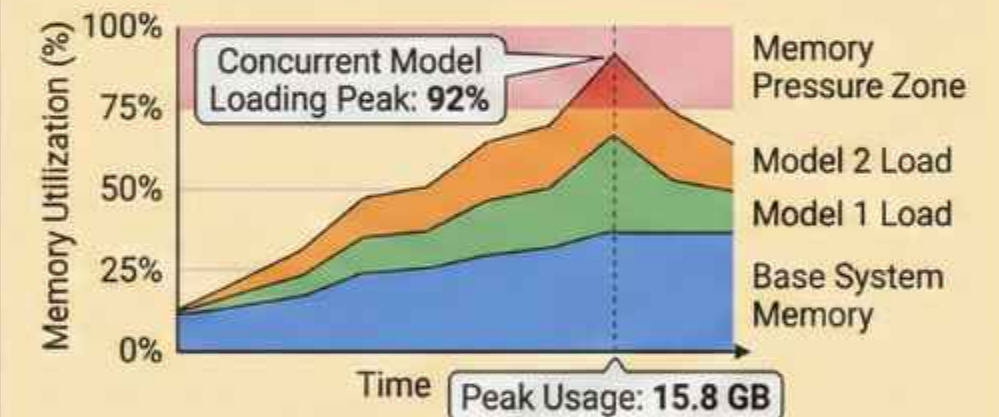
Model Quantization



Latency vs Quality



Memory Stress Test



INVESTIGATION & EXPLORATION



Memory Reduction:

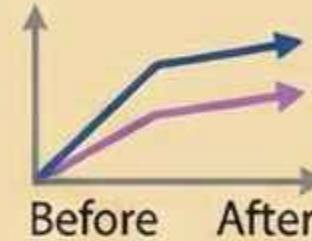
40-50% ↓

($p < 0.01$)



Latency Improvement:

25-35% ↓



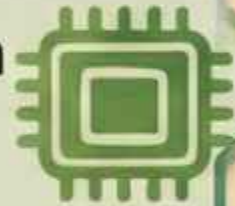
Efficiency Gains:

2x / 15% ↓



Positive Performance Conclusions

- ✓ Significant memory footprint reduction
- ✓ Accuracy remains uncompromised
- ✓ Substantial efficiency improvements on all devices



Nuanced Trade-offs & Observations

- ⚠ Latency improved, but extreme quantization
- ⚠ Minor accuracy drops in edge cases
- ⚠ Necessary trade-off for ultra-low power



Efficiency Gains & Scalability

- ▲ Double the throughput on multi-core systems
- ✓ Reduced energy consumption



Secondary Analytical Points

- 🔍 Memory bandwidth may be a bottleneck
- 🔍 Issue with larger batch sizes
- 🔍 Warrants further investigation



TOOLS & TECHNOLOGIES

Unified Technology Justification

Frameworks



PyTorch: Flexible, dynamic deep learning research & development platform.



TensorFlow: Production-ready, scalable machine learning framework.



FastAPI: High-performance, modern, async web API framework.

Models & Processing



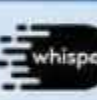
TinyLlama: Compact language model for efficient edge devices.



ResNet FER: Accurate facial expression recognition network.



SER: Speech Emotion Recognition for audio sentiment analysis.



Faster-Whisper: Fast, accurate speech-to-text transcription.

Frontend & Storage



HTML/CSS/JS: Standard web technologies for responsive user interfaces.



SQLite: Lightweight, self-contained, serverless local storage.



EdgeTTS: Low-latency, high-quality text-to-speech synthesis.

Reasoning

Async Performance
Non-blocking operations,
high throughput.



Heterogeneous Model Support
Seamless integration of
diverse AI models.

Optimized Resource Utilization
Efficient CPU/GPU usage
for edge constraints.



Value



Real-time Multimodal Interaction

Reduced Latency & Memory



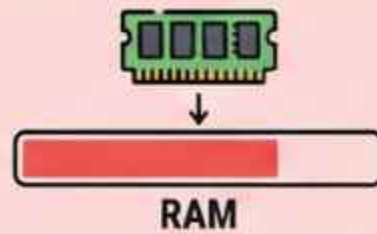
Scalable Foundation



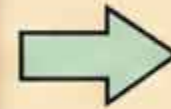
CHALLENGES & SOLUTIONS



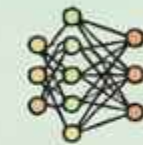
Excessive RAM usage & silent crashes



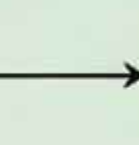
Leaks, Large Models, OOM Errors



TFLite conversion, sequential loading, garbage collection



Original Model



TFLite Conversion
(Quantization, Optimization)



Sequential Loading
(Chunking Data)

```
model = tf.lite.Interpreter(...);  
model.allocate_tensors();  
... process_chunk();  
gc.collect();
```

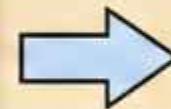
Garbage Collection
(Explicit Memory Mgmt)



Browser timeouts during LLM inference



Long processing times, default limits



Extended OS/network timeouts



OS/Browser
Settings

```
fetch(url, {  
  timeout: 60000  
});
```

Increase Timeout
Limits
(HTTP, Socket, Keep-Alive)

```
// Increase to 10 mins  
server.config.timeout = 600;
```

Backend/Server
Config
(Adjust Request Handling)

Asynchronous
Processing
(Background Tasks)



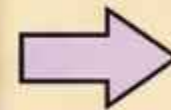
TFLite tensor shape inconsistencies



Expected Tensor
(e.g., [1, 224, 224, 3])

Actual Tensor
(e.g., [?, ?, 3])

Dynamic inputs, model conversion issues



Custom TFLite tensor wrapper with dynamic allocation



TFLite
Interpreter
Wrapper

```
wrapper = CustomInterpreter(model_path);  
wrapper.resize_tensor(input_shape);  
wrapper.invoke();  
output = wrapper.get_output();
```

Dynamic Shape
Analysis
(Inspect Input Data)

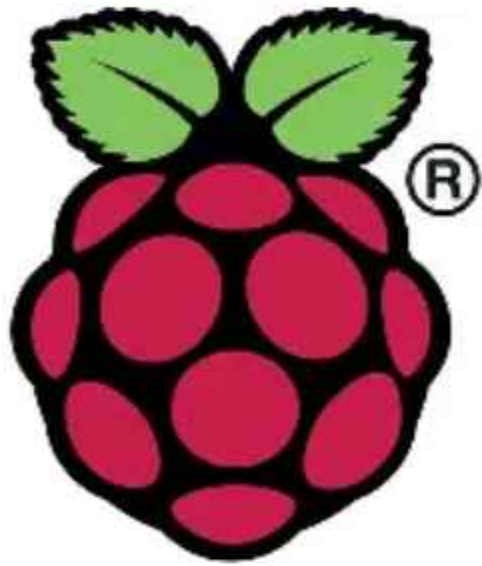
Runtime Resizing
(ResizeInputTensor)

Validated Tensor
Output
(Consistent Shapes)

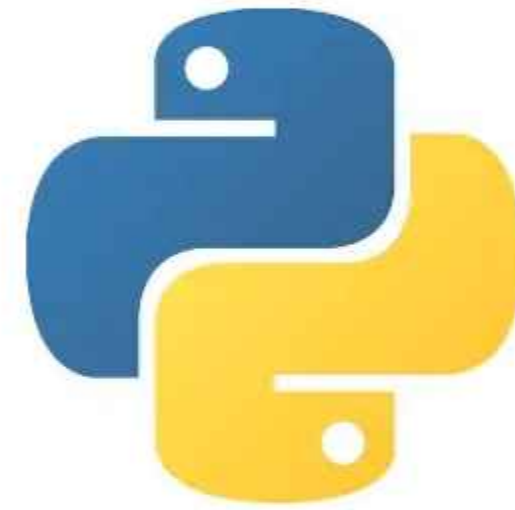
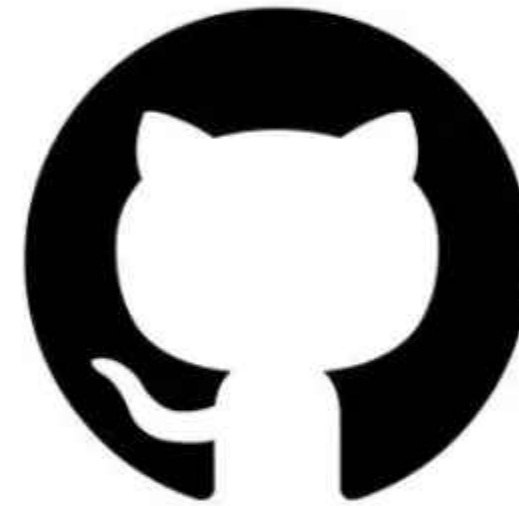
GANTT CHART



PROJECT TOOLS



Raspberry Pi



kaggle



CONCLUSION

- LLM Co-development: Enabled real-time, empathetic dialogue tuned for therapy-like sessions.
- Database Architecture: Provided long-term personalization & continuity with emotional history.
- Database-LLM Integration: Allowed context-aware, human-like conversations.
- 3D-Printed Enclosure: Delivered professional, portable, and deployable hardware.
- Overall Impact: A scalable, privacy-preserving, real-time emotional support system.



THANK YOU!

Any Questions?

TEAM STRUCTURE AND RESPONSIBILITIES

Signal Processing and Audio AI

Muhammad Taha

Speech-Emotion Modeling

Multimodal Fusion (Audio Input)

STT/TTS Refinement

Computer Vision & Visual AI

Sohaib Afzal

Facial-Emotion (Macro/Micro) Modeling

Multimodal Fusion (Facial Input)

YOLO Face Detection

Generative AI & Embedded Systems

Hashir Ashraf

LLM Fine-Tuning

Ethical & Content-Safety Filters on LLM

Raspberry Pi 5 Deployment

System Architecture & Integration

Muhammad Muhtashim

Develop Central User Database Management

Software Development

Full Pipeline Integration

LLM-FINETUNING

Design Objectives

- Empathetic, therapist-like dialogue (non-diagnostic)
- Safe mental-health support, not medical advice
- On-device & privacy-preserving
- Low-latency, resource-efficient inference
- Multilingual emotional understanding

Fine-Tuning Stages



Fine-Tuning Datasets

- DAIC-WOZ (Clinical Interviews)
- Counseichat (Professional Therapy)
- Empathetic Dialogues (Emotional Context)

Outcome

A private, on-device mental-health support assistant that provides empathetic, non-diagnostic-dialogue, priornizes user safety, preserves privacy, and avoids harmful, unethical, or misleading resporsonses.



A private, on-device mental-health support assistant that dialogus, empathetic, non-diagnostic dialogue, aissnostic, user safety, preserves privacy and avoids harmful, unalhical responses.

Model Specifications & Rationale

Architecture

Base Model:	1.1B
Layers:	22
Hidden Size:	2048
Attention Heads:	32
Context Length:	2048
Vocab Size:	32k

LoRA Config

Rank (r):	64
Alpha:	128
Dropout:	0.05
Target Modules:	7 layers
Quantization:	4.bif4
Training:	FP16

Rationale:

- TinyLLama chosen for edge deployment feasibility
- LoRA enables task adaptation without full retraining
- Quantization ensures real-time inference on low-power devices

Model Sizes & Deployment

637 MB

Base Model Size
(TinyLlama-1.1B-Chat)

208 MB

Fine-tuned Model
(with LoRA adapters)

Deployment Metrics

Inference Time: ~2.3s	~2GB RAM
Platform: Pi 5 / x86	Server: Ollama Local
Adapter Size: ~208 MB	Epochs: 5

Deployment Metrics

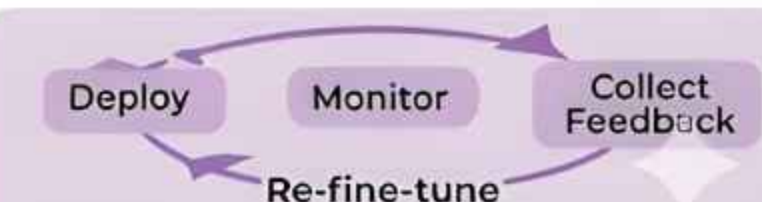
- Inference Time: ~2.3s
- Memory Usage: ~2GB RAM
- Platform: Pi 5 / x86
- Adapter Size: ~208 MB Epochs: 5

Evaluation Metrics

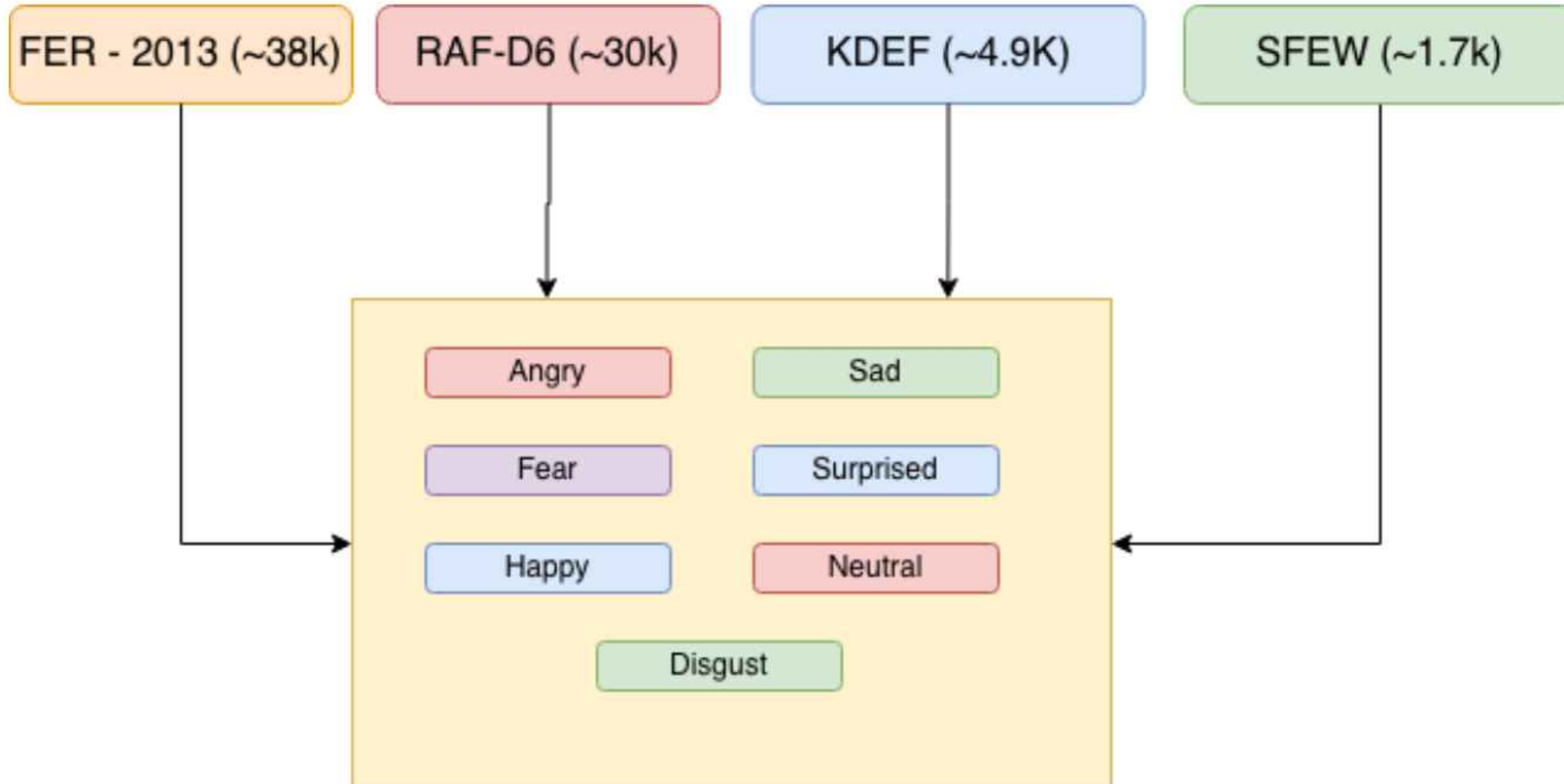
- Empathy score (human-in-the-loop evaluation)
- Response coherence & relevance
- Safety violation rate
- Hallucination & escalation checks
- Bias & toxicity screening

Safety & Governance Pipeline

- Prompt level content fients
- Rule-based crisis escalation blocking
- Retuard a reduction templates
- Dataset-level pale sanitsaican
- Continuous post-deployment monitoring



MACRO FACIAL EXPRESSION DATASET



MICRO FACIAL EXPRESSION DATASETS

Input (12.22 GB)

📁 Data Sources

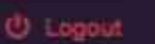
▼ 📁 Micro Facial Expressions

▶ 📁 CASME

▶ 📁 CASME II

FRONTEND DEMO

 SERENITY



 SUBJECT

Name: **muht**
Status: Online

 LOGS

[11:53:16 AM]	angry
[11:53:14 AM]	angry
[11:53:14 AM]	angry
[11:53:12 AM]	angry
[11:53:12 AM]	surprise
[11:53:10 AM]	fear
[11:53:10 AM]	fear
[11:53:08 AM]	surprise
[11:53:08 AM]	fear
[11:53:06 AM]	angry

 VISUAL CORTEX

EMOTION
angry
56%





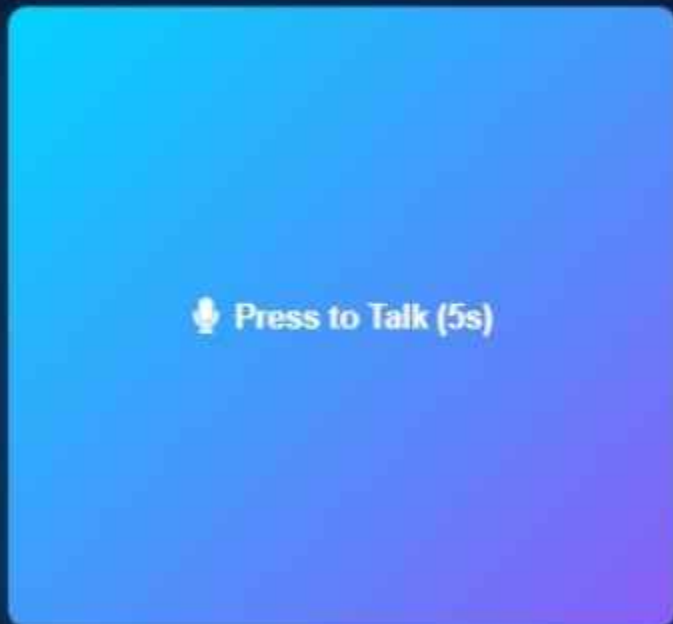


 NEURAL DIALOGUE

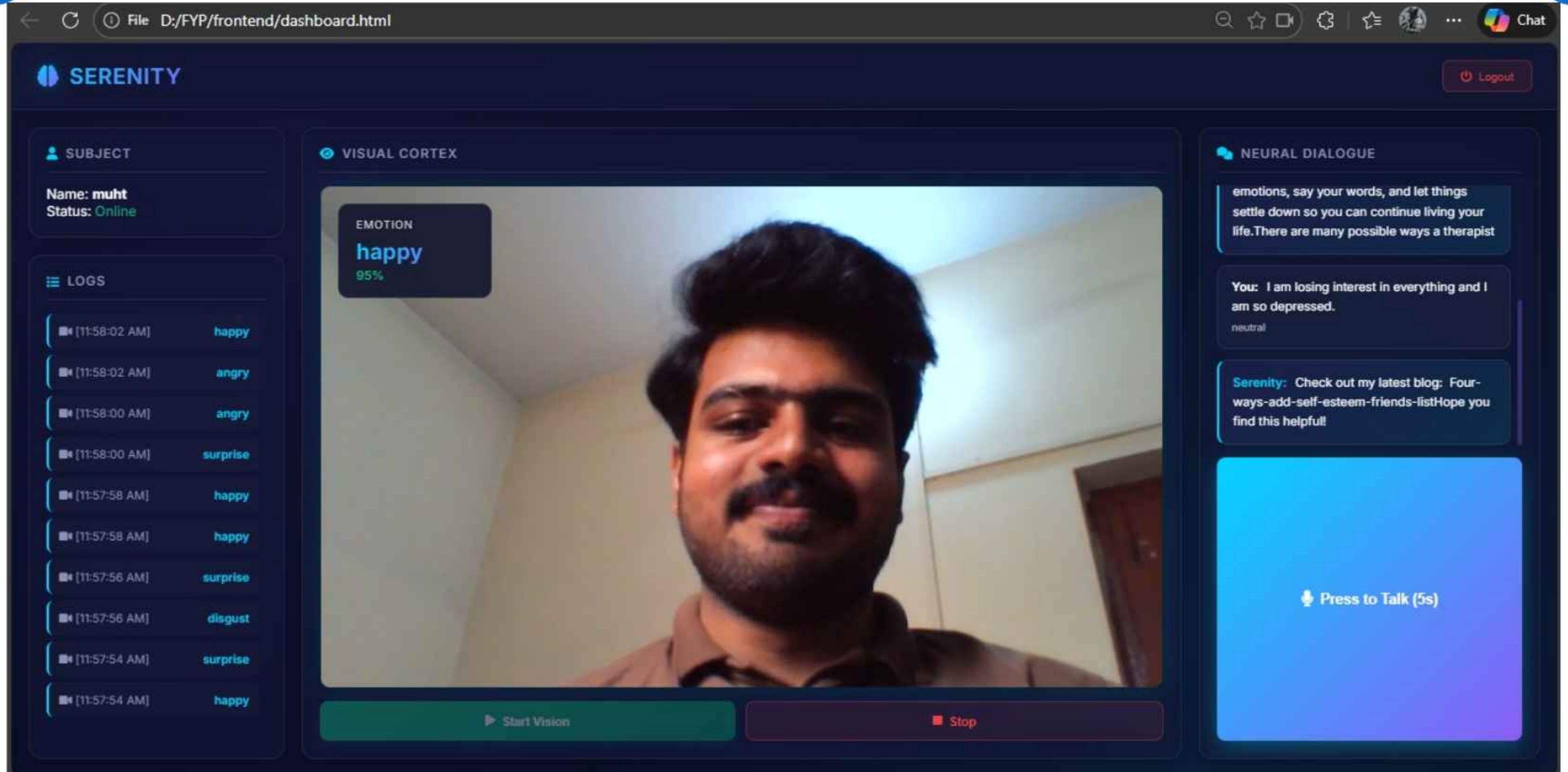
System: Ready. Press the button below to speak.

You: Can you help me?
neutral

Serenity: You are asking for help. What help would you like? Leave the situation, feel your emotions, say your words, and let things settle down so you can continue living your life. There are many possible ways a therapist



FRONTEND DEMO



FRONTEND DEMO

FileD:/FYP/frontend/dashboard.html

Chat

SERENITY

Logout

SUBJECT

Name: muht
Status: Online

LOGS

[12:07:08 PM]disgust

[12:07:06 PM]disgust

[12:07:06 PM]neutral

[12:07:04 PM]neutral

[12:07:04 PM]neutral

[12:07:02 PM]disgust

[12:07:02 PM]disgust

[12:07:00 PM]disgust

[12:07:00 PM]disgust

[12:06:58 PM]angry

VISUAL CORTEX

EMOTION

disgust

52%



Start Vision

Stop

NEURAL DIALOGUE

You: Can you give me a schedule to stay out of my fair?

neutral

Serenity: I can give you a general idea of when activities are most popular for each month of the year. The popularity of activities can also change depending on the location you are staying. For example, if you're staying in a beach

Press to Talk (5s)

FRONTEND DEMO

You: Can you give me a schedule to stay out of my fair?

neutral

Serenity: I can give you a general idea of when activities are most popular for each month of the year. The popularity of activities can also change depending on the location you are staying. For example, if you're staying in a beach

 Press to Talk (5s)

You: I worked on this project for weeks, and my boss didn't even say thank you; I feel completely invisible.

Therapist: I can sense how frustrated you're feeling. What do you think might be making this particularly challenging right now?

You: My partner has been texting their ex behind my back. Should I look through their phone tonight?

Therapist: I can understand why that would feel intense. What's been different about how you're experiencing this?

You: I honestly just don't see the point of getting out of bed most mornings.

Therapist: That must feel really exhausting. What would you like me to understand about what's been going on with you?

You: I keep pretending I'm fine because I'm terrified that if I show my real self, my friends will leave me.

Therapist: I can feel how demanding this is for you. What would you like me to know about what makes this situation unique?

You: I don't know, I just feel off today.

Therapist: I can hear how much this is affecting you. What would you like me to understand about the way you're navigating this?

You: quit